

NEUROSCIENCE

Abnormal O-glycan sialylation in the mPFC contributes to depressive-like behaviors in male mice

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Depression presents challenges in understanding its biological basis and developing effective treatments. Post-translational modifications provide critical insights into molecular mechanisms of disorders, yet their role in depression remains underexplored. Glycosylation has been particularly difficult to study due to its heterogeneous nature and analytical challenges, despite its potential to uncover previously unknown mechanisms. In this study, we demonstrate how chronic stress alters region-specific O-glycosylation across multiple brain regions, with a particular focus on the prefrontal cortex (PFC). Our findings reveal significant changes in sialylated O-glycosylation patterns mediated by *St3gal1*, a sialyltransferase essential for O-glycosylation. Notably, knockdown of *St3gal1* in nonstressed mice induced depressive-like behaviors, whereas its overexpression in stressed mice alleviated depressive symptoms, underscoring its regulatory role in stress resilience. In addition, we identify potential glycoprotein targets and downstream regulators of *St3gal1*, including neurexin 2 (NRXN2) in the medial PFC (mPFC), offering insights into molecular pathways linking O-glycosylation to depressive-like behaviors.

INTRODUCTION

Depression, a debilitating mental health disorder, remains a substantial global health challenge. Despite the availability of various antidepressants, treatment outcomes are often suboptimal due to limited efficacy, delayed onset of action, and adverse effects (1). Selective serotonin reuptake inhibitors (SSRIs) and serotonin-norepinephrine reuptake inhibitors (SNRIs) frequently fail to provide sufficient relief for many patients and are associated with side effects such as weight gain, sexual dysfunction, and an increased risk of suicidal ideation (2, 3). Emerging therapies, including psilocybin (found in magic mushrooms) and esketamine, have demonstrated promise in treatment-resistant depression but carry potential risks of abuse and other side effects (4). This underscores the necessity of identifying therapeutic targets to advance depression treatment.

Posttranslational modifications (PTMs) have recently emerged as a promising frontier in understanding brain disorders, offering insights into previously unexplored molecular mechanisms beyond genes and proteins (5, 6). Among these, glycosylation—a biochemical process involving the attachment of sugars to proteins or lipids—stands out as one of the most complex PTMs (7). Despite its critical role in protein function, stability, and key biological processes such as glucose metabolism, cell signaling, and immune responses, glycosylation remains underexplored due to its inherent heterogeneity and the significant need for sophisticated tools and methodologies (5). Emerging evidence has identified glycosylation alterations in the blood and cerebrospinal fluid (CSF) of individuals with depression

(8, 9). However, whether these alterations extend to the brain and how they contribute to depression's development remain critical unanswered questions. Addressing these gaps is essential for deepening our understanding of the molecular underpinnings of depression and uncovering therapeutic targets.

O-glycosylation, a specialized form of glycosylation in which sugars attach to the oxygen atoms on serine or threonine residues of proteins, plays a critical role in diverse biological processes, including brain function (5). This PTM, primarily occurring in the Golgi apparatus, substantially affects the function and stability of proteins essential for neural health and activity (10, 11). Mucin-type O-glycans, a major subclass of O-glycosylation, are found on numerous cell surface and secreted proteins (12–14). These glycans uniquely modulate cell recognition, adhesion, and communication with the extracellular environment, emphasizing their importance in neural processes (10, 12–14). Notably, mucin-type O-glycosylation has been implicated in mental illnesses, with evidence linking its regulatory enzymes to various neurological disorders (15, 16). For instance, microarray analyses in major depressive disorder (MDD) have revealed region-specific alterations in O-glycosylation enzymes, such as increased expression of *GALNT2* and *GALNT12* in the striatum (STR) (15). These findings suggest regionally distinct roles for O-glycosylation enzymes in the pathophysiology of MDD (5). Moreover, a cross-disorder genome-wide association study (GWAS) encompassing MDD, bipolar disorder, and schizophrenia highlighted *ST3GAL1*, a key O-glycosylation enzyme, as strongly associated with functional impairments under these conditions (16). Such findings underscore the potential role of O-glycosylation in mental health disorders.

Given these insights, glycomics, a rapidly advancing field dedicated to analyzing changes in glycan composition and distribution, presents a promising avenue for identifying biomarkers and therapeutic targets for depression. By focusing on glycosylation patterns, particularly O-glycosylation, this study aims to elucidate its role in the pathophysiology of depression. Understanding these intricate biochemical modifications and O-glycosylation-driven molecular events in the brain not only reveals critical mechanisms underlying

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the disorder but also underscores the potential of O-glycosylation in developing more effective treatment strategies for complex, multifactorial, and multisymptomatic mental disorders.

RESULTS

Chronic variable stress induces abnormal brain O-glycosylation in adult male mice

To investigate the association between depression and its effect on brain O-glycosylation, we established the depression mouse model using a typical chronic variable stress (CVS) protocol (Fig. 1A). The CVS paradigm has proven successfully in producing animal models for the study of stress-related diseases including MDD (17, 18). Novelty-suppressed feeding test (NSFT) is used as a measure of depressive-like behaviors (19). After 3 weeks of CVS, the mice significantly lost body weight and displayed anxiety-driven reduced motivation to feed, as indicated by an increased latency to approach and bite familiar chow in a novel environment during the NSFT (Fig. 1, B and C). Our protocol for generating depression models

was well established, consistently inducing depressive-like behaviors and changes in physical properties in mice, as demonstrated by three discrete sets of CVS mice (fig. S1).

We first performed brain region-specific O-glycomics to clarify the subregions exhibiting altered O-glycosylation characteristics due to chronic stress. Nine regions of brain were anatomically dissected, and O-glycans of each region were isolated and analyzed by liquid chromatography–mass spectrometry (LC-MS) and collision-induced dissociation tandem mass spectrometry (CID MS/MS) (Fig. 2). In the wild-type adult male mouse brain tissues, a total of 17 O-glycans were identified (table S1), and they were classified into mucin type O-glycans including core 1, 2, and 3 subtypes and O-mannosylated glycans, which was coincided with recent studies on O-glycomics of mouse or murine brain (20, 21). The detection frequency shown in Fig. 2A indicates whether each O-glycan was identified in each brain subregion across triplicate LC-MS analyses. Nine O-glycans were consistently detected in all nine brain regions with 100% frequency across replicates. In contrast, eight O-glycans with less than 100% detection frequency suggest that these glycans are

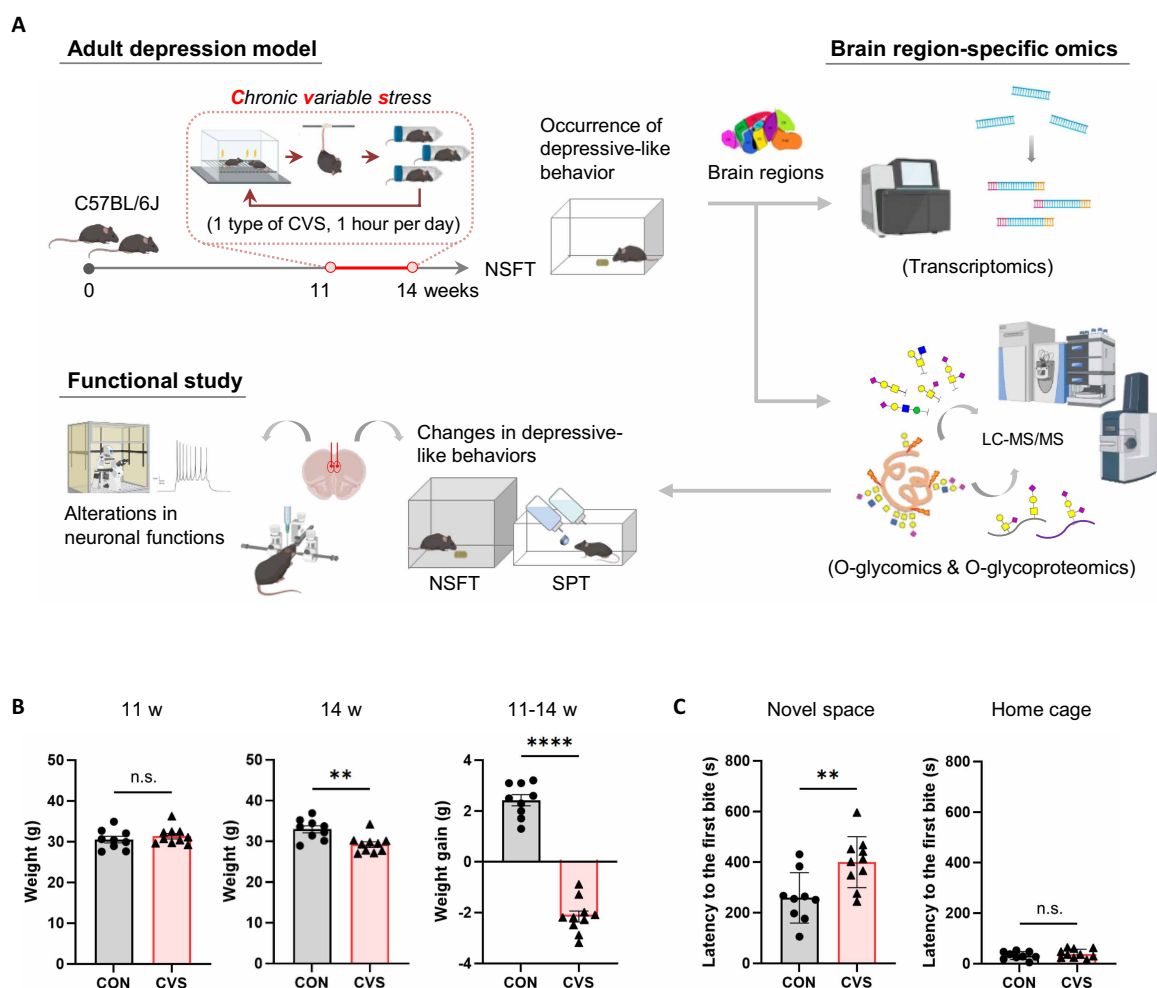


Fig. 1. Functional omics strategy for studying brain O-glycosylation in depression. (A) Schematic representation of the brain O-glycomics study of stress-induced depression in adult male mice. CVS paradigm for inducing the adult depression model is inserted in the pink dashed box. Created in BioRender. Chang, B. (2025) <https://BioRender.com/lbo2qiz>. (B) Changes in physical characteristics of adult mice after 3 weeks of CVS ($n = 9$ CON; $n = 10$ CVS), presented as body weights before (11 weeks) and after CVS (14 weeks) and weight gain (11 to 14 weeks) following CVS. w, weeks. (C) Results from the NSFT (left) and feeding test in the home cage (right) of adult male mice exposed to CVS. Statistical analysis was performed using a two-tailed unpaired t test. ** $P < 0.01$; **** $P < 0.0001$. n.s., not significant. Error bars represent SEM.

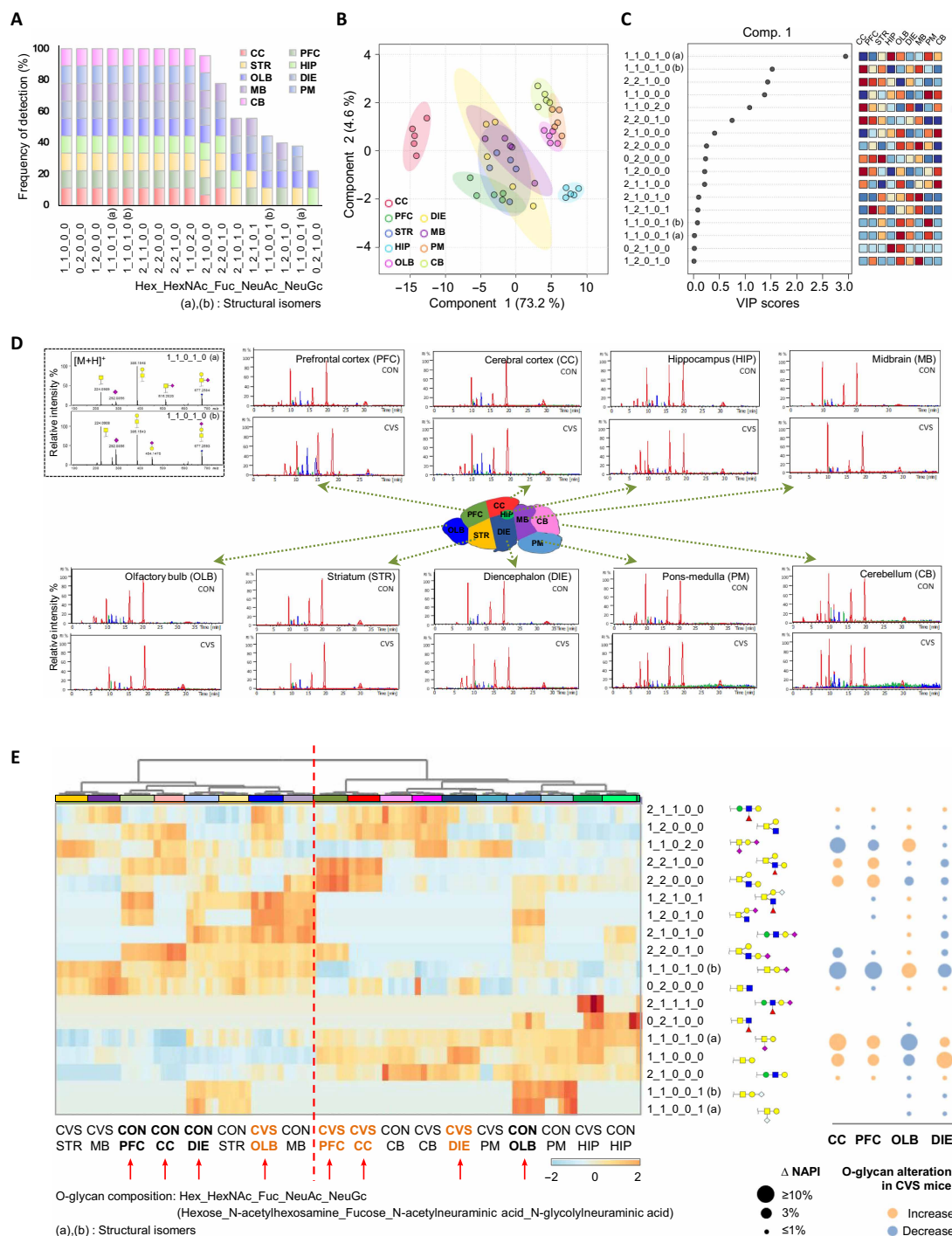


Fig. 2. Sialylated O-glycosylation patterns in the mouse brain vary across regions and depression status. (A) Frequency of detection (%) of O-glycan compositions according to the nine brain regions of adult male mouse. Each brain region from two mice was pooled for O-glycomic analysis ($n = 5$ pooled samples per region), and each sample was analyzed in triplicate by LC-MS. (B) PLS-DA plot of region-specific O-glycosylation. (C) VIP score for comp. 1. (D) EICs of O-glycans obtained from nine brain regions (top, CON; bottom, CVS) in LC-MS analysis. Chromatographic peaks were differently colored according to O-glycosylation types including core 1-type (red), core 2- and core 3-type (blue), and O-mannosylation (green). The x axis is the retention time (minutes). The y axis represents the relative intensity (RI %) with the highest peak assigned as intensity 100%. CID MS/MS spectra in the dashed box exhibited structural isomers of the sialylated O-glycan of m/z 677.25 $[M + H]^+$. Monosaccharide symbols of O-glycans represent galactose (Gal, yellow circle), N-acetylgalactosamine (GalNAc, yellow square), N-acetylglucosamine (GlcNAc, blue square), fucose (Fuc, red triangle), and NeuNAc (purple diamond). (E) Hierarchical clustering heatmap of O-glycosylation in nine brain regions of control and CVS mice ($n = 5$ per group). The dashed line indicates the two main clusters (left). Bubble plot of the extent of O-glycosylation changes in the brain regions (indicated by red arrows) grouped separately in the heatmap (right). NAPI (%) means normalized absolute peak intensity. The sum of chromatographic peak area of all O-glycan ions in each region is converted to 100%. Δ NAPI indicates increased or decreased amount of each O-glycan in brain regions after CVS.

absent in some brain regions. Notably, core 1–derived O-glycans were the most common and frequently observed across the nine brain regions of adult male mice (Fig. 2A). On the other hand, the relative quantities of O-glycan components, profiled using the peak areas of O-glycan ions in LC-MS analysis, showed significant differences across the nine brain regions, indicating region-specific O-glycosylation characteristics in the mouse brain, which were demonstrated in this study (Fig. 2, B to D). Specifically, various O-glycan compositions and their relative abundances were identified on the basis of ion peaks and peak areas in extracted ion chromatograms (EICs) obtained from LC-MS analysis (Fig. 2D). The detailed structures of these O-glycans were further characterized by CID MS/MS analysis, as illustrated in the black dashed box in Fig. 2D. By combining chromatographic separation with MS/MS, we successfully distinguished structural isomers of O-glycans and pinpointed the specific locations of *N*-acetylneuraminic acid (NeuNAc) moieties. This is exemplified by the isomers of (GalNAc₁-NeuNAc₁)Gal₁ and GalNAc₁(Gal₁-NeuNAc₁) [1_1_0_1_0 (a) and 1_1_0_1_0 (b)], highlighted within the black dashed box in Fig. 2D. Using this validated method, the cerebral cortex (CC) and hippocampus (HIP) of wild-type adult male mouse brains showed markedly distinct O-glycosylation patterns in the partial least squares discriminant analysis (PLS-DA) plot (Fig. 2B). In addition, variable importance in projection (VIP) analysis identified key O-glycans that significantly contributed to the observed PLS-DA score plot (Fig. 2, B and C).

We further analyzed the expression of individual O-glycans in mice subjected to CVS using hierarchical clustering and observed significant alterations in O-glycosylation patterns within the CC, prefrontal cortex (PFC), olfactory bulb (OLB), and diencephalon (DIE). These changes were prominently visualized through heatmap analysis (Fig. 2E and table S2). Sialylated O-glycans bearing NeuNAc residues, which account for most brain O-glycosylation, were prominently affected by chronic and unavoidable stress (Fig. 2E and fig. S2). Notably, core 1–type O-glycans including galactose-linked NeuNAc (Gal-NeuNAc) were significantly reduced in the CC, PFC, and DIE of CVS mice, whereas the OLB of CVS mice revealed the sialylated O-glycosylation pattern opposite to the other three regions (Fig. 2E and fig. S2). Although the CC region also showed a significant reduction in core 1–type O-glycans including Gal-NeuNAc, this study focused on the PFC as it is known to be substantially affected and has been extensively studied for depression symptoms in both rodent models and human patients (22, 23). Overall, we found significant alterations in core 1–type O-glycans containing the Gal-NeuNAc moiety in the PFC, leading us to examine the possible role of O-glycan sialylation in depression. We hypothesized that chronic stress would affect a substantial reduction in O-glycan sialylation in the brain and subsequently be associated with depressive-like behaviors.

A notable change in the O-glycan sialylation gene, *St3gal1*, is observed in the PFC of a depression model in male mice

O-glycosylation signifies the enzymatic process of adding or removing carbohydrates, known as O-glycans, on proteins (24). We sought to determine whether differential O-glycosylation in the brains of CVS mice originates at the gene level of the enzymes involved in O-glycosylation synthesis. We initially performed bulk RNA sequencing (RNA-seq) on the PFC of CON and CVS mice. In total, 761 genes were differentially expressed in CVS versus CON mice (*P* value < 0.05). Differentially expressed genes (DEGs) in the PFC

were distinctively grouped according to stress state of mice (*n* = 3 CON and *n* = 3 CVS mice) (Fig. 3A).

According to the KEGG (Kyoto Encyclopedia of Genes and Genomes) pathway analysis, DEGs were mainly enriched in the pathways related to axonal guidance and brain diseases such as Alzheimer's disease (AD), Parkinson's disease (PD), and Huntington's disease (HD) (Fig. 3B and table S3). These findings are similar to previous transcriptomics studies of patients with MDD and animal models of depression (25–27). In addition, among the DEGs, the top 10 most significantly up-regulated and down-regulated genes include *Plin4*, *Fkbp5*, and *Irf7*, which have been previously reported to be associated with depressive symptoms, supporting the validity of our RNA-seq results (Fig. 3C and table S4) (28, 29).

In terms of glycosylation, we identified alterations in genes responsible for biosynthesis of various glycosylations such as N- and O-glycans and proteoglycans in the KEGG pathway analysis (Fig. 3B). RNA-seq data of the PFC in CVS mice showed significant changes in several O-glycosylation-related genes, including *St3gal1*, *Gcnt4*, *Fut9*, and *B3galt5* (Fig. 3C and table S4). To clarify the DEGs in PFC transcriptomics of CVS mice, we reexamined the four O-glycosylation-related genes by quantitative real-time reverse transcription polymerase chain reaction (RT-qPCR) (Fig. 3F and fig. S3). *St3gal1* and *Gcnt4*, glycosyltransferases involved solely in O-glycosylation synthesis, were significantly changed in the PFC. These changes were not observed in the HIP, another core brain region associated with depressive-like behavior, suggesting that chronic stress induces brain region-specific changes in O-glycosylation (fig. S4). When further exploring sialylation-related genes, a total of 4 sialidases and 19 sialyltransferases were identified in the PFC (Fig. 3D and table S5). Among these, only one sialyltransferase, *St3gal1*, was expressed significantly lower in the CVS group. In the four brain regions exhibiting significant O-glycosylation changes in CVS mice, *St3gal1* expression was down-regulated in the CC and PFC but up-regulated in the OLB, aligning with the quantitative profiling of *St3gal1*-associated sialylated O-glycans (Fig. 3, E to H, and table S6).

We further investigated changes in sialylation of brain O-glycans in stress-exposed female mice. The same CVS protocol applied to adult male mice was used to generate an adult female model exhibiting depressive-like behaviors (fig. S5). After experiencing stressful events for 3 weeks, adult female mice also lost body weight and displayed increased latency to feed in the NSFT as an evaluation of depression-related behaviors with anxiety and motivational deficits (fig. S5). However, no changes in *St3gal1* gene expression were observed in the brain regions, CC, PFC, HIP, and OLB of female CVS mice, suggesting that chronic stress-induced O-glycosylation changes in the brain are not consistent between male and female mice (fig. S5).

Inhibition of *St3gal1* expression in the mPFC induces depressive-like behaviors in normal male mice

As reported in a previous study on the global genotyping of patients with mental disorders, *ST3GAL1* was identified as one of the target genes associated with MDD, bipolar disorder, and schizophrenia (16). In this study, we aimed to uncover the causal relationship between *St3gal1*-related O-glycosylation and depressive-like behaviors. To explore the impact of inhibited *St3gal1* in brain, we generated the mice with targeted knockdown of *St3gal1* in the medial PFC (mPFC) by inducing *St3gal1* short hairpin RNA (shRNA) using adeno-associated virus (AAV) vectors and their behaviors were observed under a nonstressed condition. *ST3GAL1* is known to be mainly

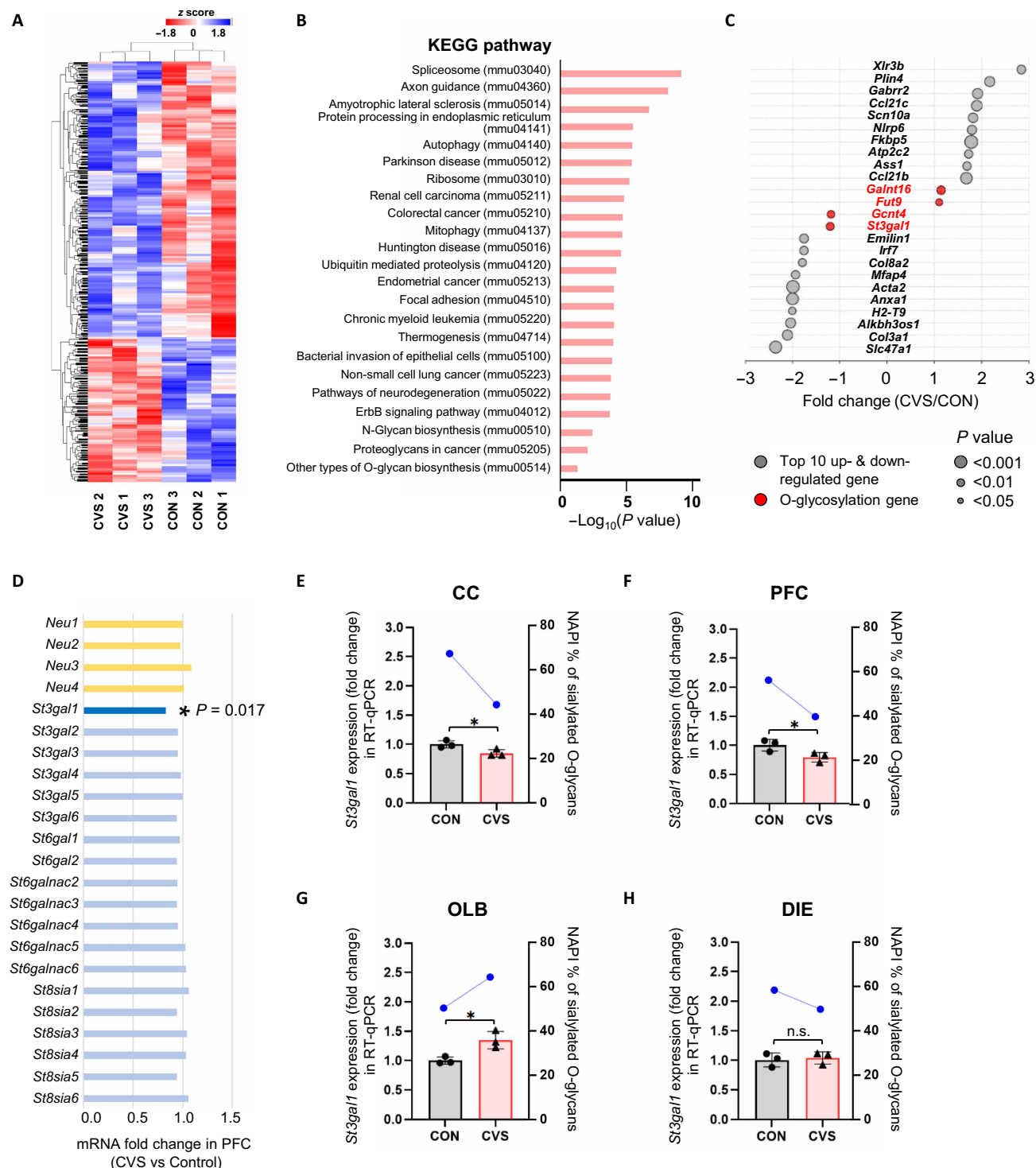


Fig. 3. O-glycosylation-specific sialylation gene, *St3gal1*, is altered in the PFC of adult depression mice. (A) Heatmap of PFC DEGs of adult mice ($n = 3$ CON; $n = 3$ CVS) (P value < 0.05). (B) KEGG pathways of the DEGs altered in PFC after CVS (top 20 and glycosylation-related pathways). (C) Top 10 up-regulated and down-regulated DEGs (black) and O-glycosylation-related genes (red) of PFC (CVS versus CON mice). (D) Fold changes in sialylation-associated genes, sialyltransferases and sialidases, in RNA-seq of PFC from CVS versus CON mice. (E to H) Correlation between alterations of *St3gal1* expression and sialylated O-glycans synthesized by ST3GAL1. (E) CC, (F) PFC, (G) OLB, and (H) DIE are the four regions clearly separated by control and CVS group in the heatmap analysis of brain O-glycomics shown in Fig. 2E. The bar indicates the fold change of *St3gal1* in RT-qPCR analysis (y axis on the left). Two-tailed unpaired t test. $*P < 0.05$. Error bars represent SEM. The dot shows NAPI % of sialylated O-glycans synthesized by ST3GAL1 (y axis on the right).

expressed in neurons compared to other types of cells such as astrocytes and oligodendrocytes (30). Therefore, we first tested the knockdown efficiency of AAV-shRNA vectors containing three different shRNA sequences of *St3gal1* as candidates in primary mouse cortical neurons (fig. S6). We selected the most efficient shRNA (shRNA1), which down-regulated *St3gal1* by 60% in neurons. AAV vectors containing either *St3gal1*-shRNA (*shSt3gal1*) or scramble-shRNA (Scramble, SCR) were bilaterally injected into the mPFC of 11-week-old male mice. We first confirmed that the transfected cells remained healthy as indicated by 4',6-diamidino-2-phenylindole (DAPI) staining (Fig. 4B). To determine the efficiency of viral transfection in adult mice, the mRNA expression of *St3gal1* was measured in the mPFC 10 days after injection of AAV-shRNA vectors. The mPFC in *St3gal1* knockdown mice showed a significant reduction in *St3gal1* gene expression at both the mRNA (Fig. 4C) and protein levels (fig. S7), compared to the mPFC in control shRNA-injected mice. Ten days postinjection, their depressive-like behaviors were sequentially evaluated in the following order: NSFT, open field test (OFT), tail suspension test (TST), and sucrose preference test (SPT). In the NSFT, we found that *St3gal1* knockdown mice showed an increased latency to feed in the novel environment compared to control mice, with no significant changes observed in the home cage (Fig. 4D). In the OFT, the knockdown mice showed no significant difference in anxiety compared to the controls, although a trend was observed. This was measured by the time spent or distance traveled in the center area of the arena during the first 5 min (Fig. 4D and fig. S8). There were no significant differences in the general locomotion, measured by total distance traveled in 30 min,

between the knockdown and control mice, indicating normal locomotion in the knockdown mice (fig. S8). In addition, no differences were observed in the TST (Fig. 4D). In the SPT to evaluate anhedonia, the *St3gal1* knockdown mice showed a significant decrease in preference for sucrose, revealing a decreased ability to experience pleasure (anhedonia) (Fig. 4D). Together, although not in all aspects of depressive-like behaviors, we clearly determined that the down-regulation of the O-glycan sialylation gene *St3gal1* leads to the development of depressive-like behaviors, including novelty-induced feeding suppression and anhedonia-like symptoms, in healthy adult male mice, even in the absence of stressful events.

Overexpression of *St3gal1* can improve depressive-like symptoms in stress-exposed male mice

To test whether modulating *St3gal1* in brain could improve the symptoms of an adult depression model, we performed the overexpression of *St3gal1* (*St3gal1*-OE) targeting to neurons in the mPFC of CVS mice. As depicted in Fig. 5A, male mice were exposed to 3 weeks of CVS and then injected with either control enhanced green fluorescent protein (eGFP) viral vectors (CVS.GFP mice) or *St3gal1*-OE viral vectors (CVS.*St3gal1*-GFP mice) into the mPFC. In addition, an age-matched male control group, kept in the home cage without exposure to CVS and injected with a control virus (CON.GFP), was included. After 10 days, all mouse groups were subjected to depressive-like behavioral tests such as the NSFT, OFT, TST, and SPT.

The expression efficiency of the control and *St3gal1*-OE viral vectors, driven by the human synapsin 1 (*hSyn1*) promoter, was validated in primary mouse cortical neurons. A significant increase in

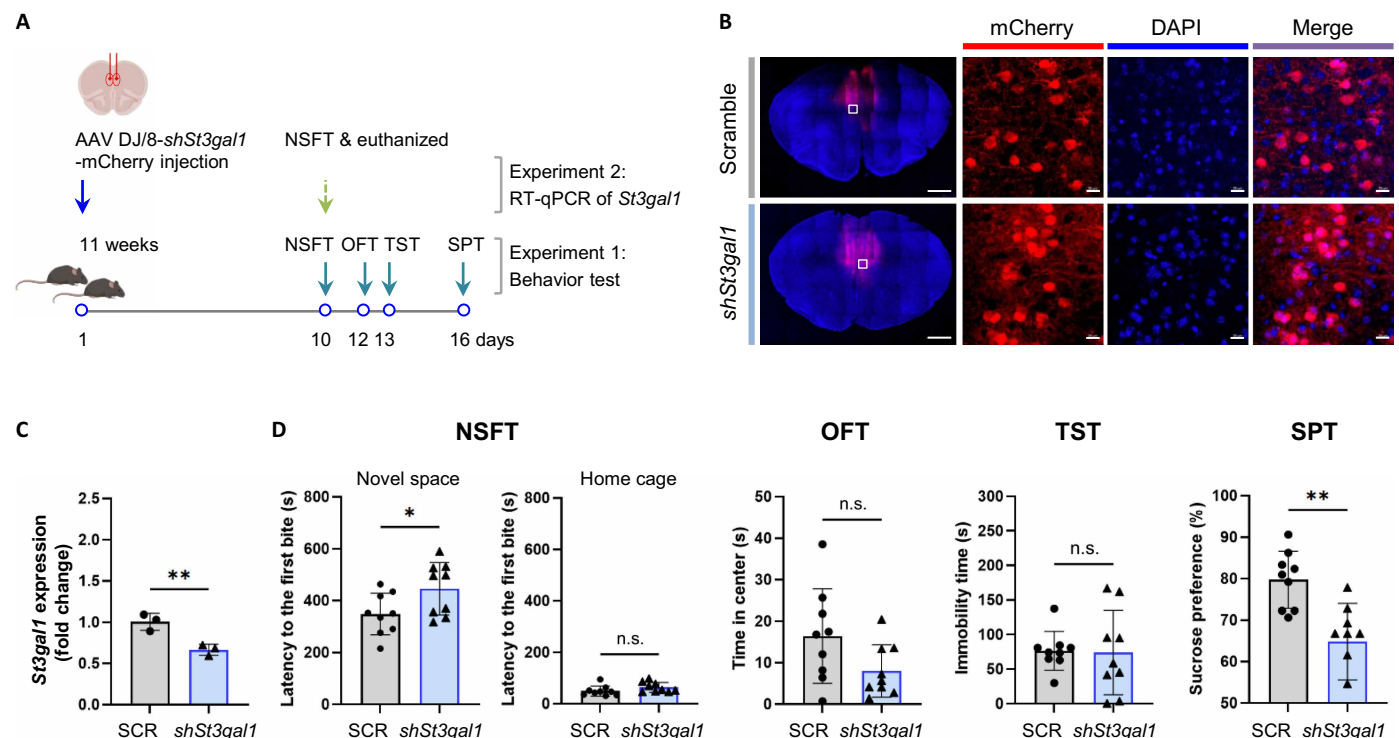


Fig. 4. mPFC-specific knockdown of *St3gal1* causes depressive-like behaviors in normal male mice. (A) Streamlined outline of the *St3gal1* knockdown experiment. Created in BioRender. Chang, B. (2025) <https://BioRender.com/w97m7qg>. (B) Representative fluorescence image of brain slices of scramble (SCR) and *St3gal1* knockdown (*shSt3gal1*) mice. Scale bars, 100 μ m (whole slice images) and 20 μ m (zoomed images). (C) RT-qPCR of *St3gal1* in the PFC at 10 days after injection of AAV-scramble or AAV-*shSt3gal1* vectors. (D) Depressive-like behavior tests, NSFT, OFT, TST, and SPT, of SCR ($n = 9$) and *shSt3gal1* ($n = 9$) mice. Two-tailed unpaired t test. * $P < 0.05$; ** $P < 0.01$. n.s., not significant. Error bars represent SEM.

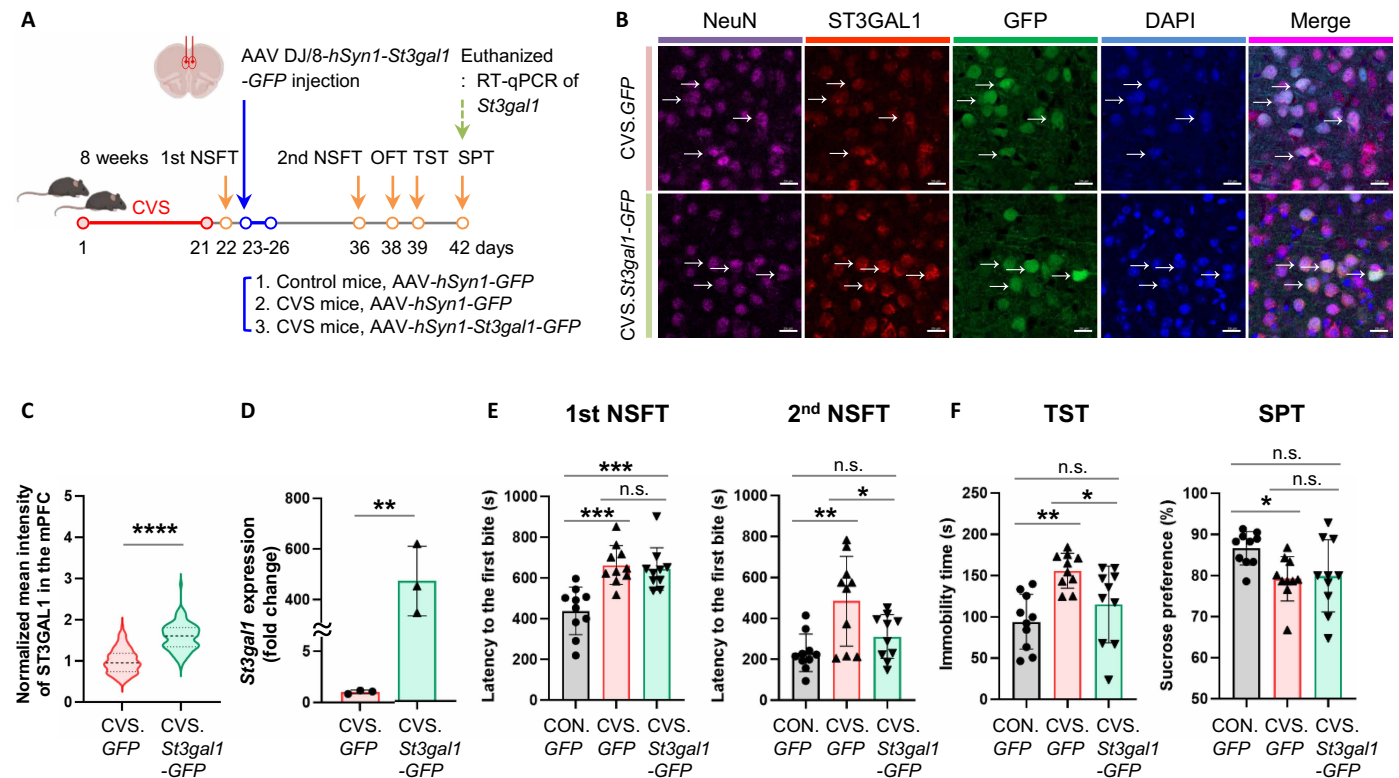


Fig. 5. Overexpression of neuronal *St3gal1* in the mPFC reverses depressive-like symptoms in male CVS mice. (A) Streamlined outline of *St3gal1* overexpression in the mPFC using AAV-hSyn1-*St3gal1*-GFP (*St3gal1*.OE) and AAV-hSyn1-GFP (control) vectors. Created in BioRender. Chang, B. (2025) <https://BioRender.com/rqwdjzj>. (B) Representative images of immunofluorescence staining of ST3GAL1 (red) and NeuN (purple) with GFP (green) in CVS.GFP (treated by control vectors) and CVS.St3gal1-GFP (treated by *St3gal1*.OE vectors) mice. Scale bars, 20 μ m. (C) Normalized mean intensity of ST3GAL1 expression in the mPFC of CVS.GFP ($n = 102$ cells) and CVS.St3gal1-GFP ($n = 102$ cells) mice. The MFI of each cell was normalized to the average MFI of cells in CVS.GFP mice. (D) RT-qPCR of *St3gal1* in the mPFC at 10 days after injection of *St3gal1*.OE or control viral vectors. Two-tailed unpaired *t* test. $**P < 0.01$; $****P < 0.0001$. (E) NSFT of CON.GFP ($n = 10$), CVS.GFP ($n = 10$), and CVS.St3gal1-GFP ($n = 10$) mouse group before (left) and after (right) injection of AAV vectors. (F) TST (left) and SPT (right) of all mouse groups after injection of AAV vectors. One-way ANOVA with Bonferroni multiple comparisons test. $*P < 0.05$; $**P < 0.01$; $***P < 0.001$. n.s., not significant. Error bars represent SEM.

the expression of *St3gal1* was observed in neurons infected with the *St3gal1*-OE viral vectors, representing a more than 70-fold change compared to the control viral vectors (fig. S9). The validated control or *St3gal1*-OE viral vectors were injected bilaterally in the mPFC of adult mice exposed to the CVS and the viral vector expression was confirmed in fluorescence images of coronal brain slices (Fig. 5A and fig. S9). Ten days postinjection, ST3GAL1 was predominantly observed in NeuN-positive cells. Enhanced ST3GAL1 protein levels in the mPFC of mice injected with the *St3gal1*-OE vector, compared to those injected with the control virus, were confirmed by an increase in mean fluorescence intensity (MFI) per cell (Fig. 5, B and C). We also determined a significant increase in *St3gal1* mRNA in the mPFC of CVS.St3gal1-GFP mice than CVS.GFP mice by RT-qPCR analysis (Fig. 5D). To verify the behavioral effect of *St3gal1* overexpression in CVS mice, we performed the NSFT before and after the virus injection (Fig. 5, A and E). For the first NSFT (1st NSFT), two groups of CVS mice showed a significant increase in the latency to the first bite, indicating that both groups exhibited depressive-like behaviors to a similar extent. In the second NSFT (2nd NSFT), the CVS.St3gal1-GFP mice group showed a significantly reduced latency to the first bite compared to CVS.GFP mice group, suggesting that, although the CVS.GFP mice exhibited elevated levels

of anxiety and reduced motivational capacity associated with depression, this symptom was significantly improved in CVS.St3gal1-GFP mice (Fig. 5E). Although *St3gal1*-OE did not rescue the anxiety in the OFT or anhedonic behavior in the SPT, it did significantly improve passive stress-coping behaviors in the TST (Fig. 5F and fig. S10). Although immobility time in the TST is widely used as an index of depressive-like behavior following CVS, potential confounding effects of habituation, sensitization, or procedural learning cannot be entirely excluded (17, 31). To address this concern, we compared immobility time between CVS.St3gal1-GFP mice and CON.GFP mice that had not been previously exposed to the TST. We observed no significant difference between the two groups (Fig. 5F), suggesting that such confounding factors are unlikely to substantially influence the interpretation of the antidepressant-like effects observed with *St3gal1* overexpression. As a separate experiment, we examined the effects of *St3gal1* overexpression in control mice (CON.St3gal1-GFP) on depressive-like behaviors (fig. S11). We observed no significant changes in baseline anxiety-like or depression-like behaviors in the OFT, NSFT, TST, or SPT, suggesting that elevated *St3gal1* levels do not influence behavior under nonstressed conditions. Together, these findings support the notion that reduced *St3gal1* expression may be specifically linked to the emergence of

depressive-like behaviors. In this context, restoring *St3gal1* expression in stressed animals appears to be critical for reversing these behavioral abnormalities.

To further investigate the role of *St3gal1* expression in the recovery of depressive symptoms, we administered the well-known antidepressant ketamine and examined *St3gal1* expression in the mPFC (fig. S12). Notably, ketamine treatment (CVS.Ketamine) significantly reduced the latency to feed in a novel environment compared to the saline-treated CVS group (CVS.Saline) (fig. S12). Moreover, the reduction in *St3gal1* expression observed in the CVS.Saline group compared to the control group was not present in the CVS.Ketamine group, suggesting its potential involvement in the antidepressant response (fig. S12). The effect of ketamine on nonstressed mice in the NSFT, compared to the saline group, was not significant, although a trend toward reduced latency in the second NSFT was observed. In addition, no significant change in *St3gal1* expression was found between the two groups, suggesting that decreased *St3gal1* expression in the mPFC is closely associated with depressive-like behaviors (fig. S12). We also did not observe any changes in the latency to the first bite across repeated measures in different contexts (fig. S12, B and E), supporting minimal learning or habituation effects when the NSFT is conducted in distinct environments. The significant reduction in latency observed in Fig. 5E in the CON.GFP group is likely attributable to surgical effects between the first and second NSFT sessions. In summary, the recovery of *St3gal1* expression in the mPFC of CVS mice could attenuate their depression symptoms, ultimately supporting that the O-glycan sialylation is crucial for controlling depressive-like behaviors caused by chronic stress.

ST3GAL1-driven O-glycoprotein targets and associated proteins are revealed in the mPFC of CVS male mice

To identify the underlying mechanisms by which ST3GAL1 in the mPFC affects specific O-glycoproteins and regulates depressive-like behaviors, we performed the glycoproteomic analysis with a focus on sialylated O-glycosylation (Fig. 6). As depicted in Fig. 6A, normal and CVS male mice were prepared by injecting control GFP or *St3gal1*-OE viral vectors into their mPFC. Depressive symptoms were assessed using the NSFT. We selectively isolated sialylated O-glycopeptides from the mPFC of mouse brains using their high binding affinity to titanium dioxide (TiO₂), enabling the identification of proteins bearing sialylated O-glycans in each mouse group. A total of 31 sialylated O-glycopeptides and 24 O-glycoproteins were determined through LC-higher-energy collision dissociation (HCD)-MS/MS analysis (Fig. 6B and table S7). In all three groups, eight proteins with sialylated O-glycan components at specific O-glycosylation sites were consistently revealed (Fig. 6B). Five sialylated O-glycoproteins were uniquely detected in both the CON.GFP and CVS.*St3gal1*-GFP groups but were absent in CVS.GFP mice (Fig. 6B). As demonstrated in Fig. 6 (B and C), we identified O-glycan sialylation at the threonine-1584 (T1584) site of the NRXN2 protein under the condition of *St3gal1* overexpression, alongside other O-glycoproteins, including DNER (T159), OMGP (T338), CSPG-5 (S304), and LRRK2 (T776). This suggests that O-glycan sialylation of these specific proteins may be diminished during chronic stress exposure and restored via neuron-targeted *St3gal1* overexpression.

In addition, we characterized the up-regulated and down-regulated proteins in the mPFC to identify molecular networks affected by *St3gal1* overexpression. The ST3GAL1 protein was significantly up-regulated in the mPFC of CVS.*St3gal1*-GFP mice, confirming

successful overexpression (Fig. 6D). In the CVS.GFP group compared to the CON.GFP group, we identified 10 up-regulated and 12 down-regulated differentially expressed proteins (DEPs) (table S8). In the CVS.*St3gal1*-GFP group compared to the CVS.GFP group, we observed 225 up-regulated and 163 down-regulated DEPs (Fig. 6, D and E, and table S8). Among these, five DEPs displayed reversed expression patterns with *St3gal1* overexpression (Fig. 6, D and E). Specifically, MERTK and MPDU1 were significantly up-regulated in the CVS.GFP group compared to the control but were significantly down-regulated in the CVS.*St3gal1*-GFP group compared to the CVS.GFP group. Conversely, MYT1, BCAS3, and H3F3A were significantly down-regulated in the CVS.GFP group compared to the control but were significantly up-regulated in the CVS.*St3gal1*-GFP group compared to the CVS.GFP group. These ST3GAL1-mediated changes highlight these proteins as potential downstream regulators linked to chronic stress-induced depressive-like behaviors (Fig. 6, D and E).

Gene Ontology (GO) analysis of DEPs in the mPFC of CVS.*St3gal1*-GFP mice compared to CVS.GFP mice provided key functional insights. DEPs were enriched in diverse biological processes such as “metabolic processes” and “localization,” molecular functions including “catalytic activity” and “protein binding,” and cellular components such as the “cytoplasm,” “membrane,” and “synapse.” These findings underscore the role of *St3gal1* overexpression-mediated DEPs in regulating metabolic processes and their functional contributions to various cellular compartments in the mPFC of stressed mice (fig. S13 and table S9). Protein-protein interaction network analysis of ST3GAL1-associated proteins, including sialylated O-glycoprotein targets (NRXN2, DNER, OMGP, CSPG5, and LRRK2) and reversibly expressed proteins (MERTK, MPDU1, MYT1, BCAS3, and H3F3A), identified three distinct clusters: Cluster 1 (dermatan sulfate biosynthesis), Cluster 2 (neurexins and neuroligins), and Cluster 3 (aspartic-type endopeptidase activity) (fig. S14). The clusters were intermingled, suggesting functional interactions between these protein groups. Among the potential ST3GAL1 effector proteins, NRXN2 exhibited the highest connectivity within the network, suggesting a central role in mediating interactions among associated proteins and highlighting NRXN2 as a key node within this protein network. These findings underscore the potential roles of ST3GAL1 and its downstream effectors in molecular pathways related to depressive-like behaviors, offering a foundation for further mechanistic studies (fig. S14).

To pinpoint molecular changes that are both stress-sensitive and *St3gal1*-responsive, we first identified glycoproteins in which O-glycan sialylation was absent in CVS.GFP mice compared to CON.GFP controls and then determined which of these alterations reappeared following *St3gal1* overexpression (CVS.*St3gal1*-GFP versus CVS.GFP). Such a stepwise and rigorous approach highlighted five bidirectionally regulated candidates—NRXN2, DNER, OMGP, CSPG5, and LRRK2—that are likely direct targets of ST3GAL1-mediated sialylation. However, to draw more definitive conclusions, future incorporation of the CON.*St3gal1*-GFP group into proteomic and glycoproteomic analyses will be valuable to determine whether the identified glycosylation signatures are specific to the chronically stressed state and dependent on *St3gal1* overexpression.

Given that the identified potential ST3GAL1 effector proteins, including NRXN2, DNER, CSPG5, OMGP, LRRK2, and MERTK, are associated with synapse formation and function, we examined the impact of *St3gal1* overexpression on the electrophysiological

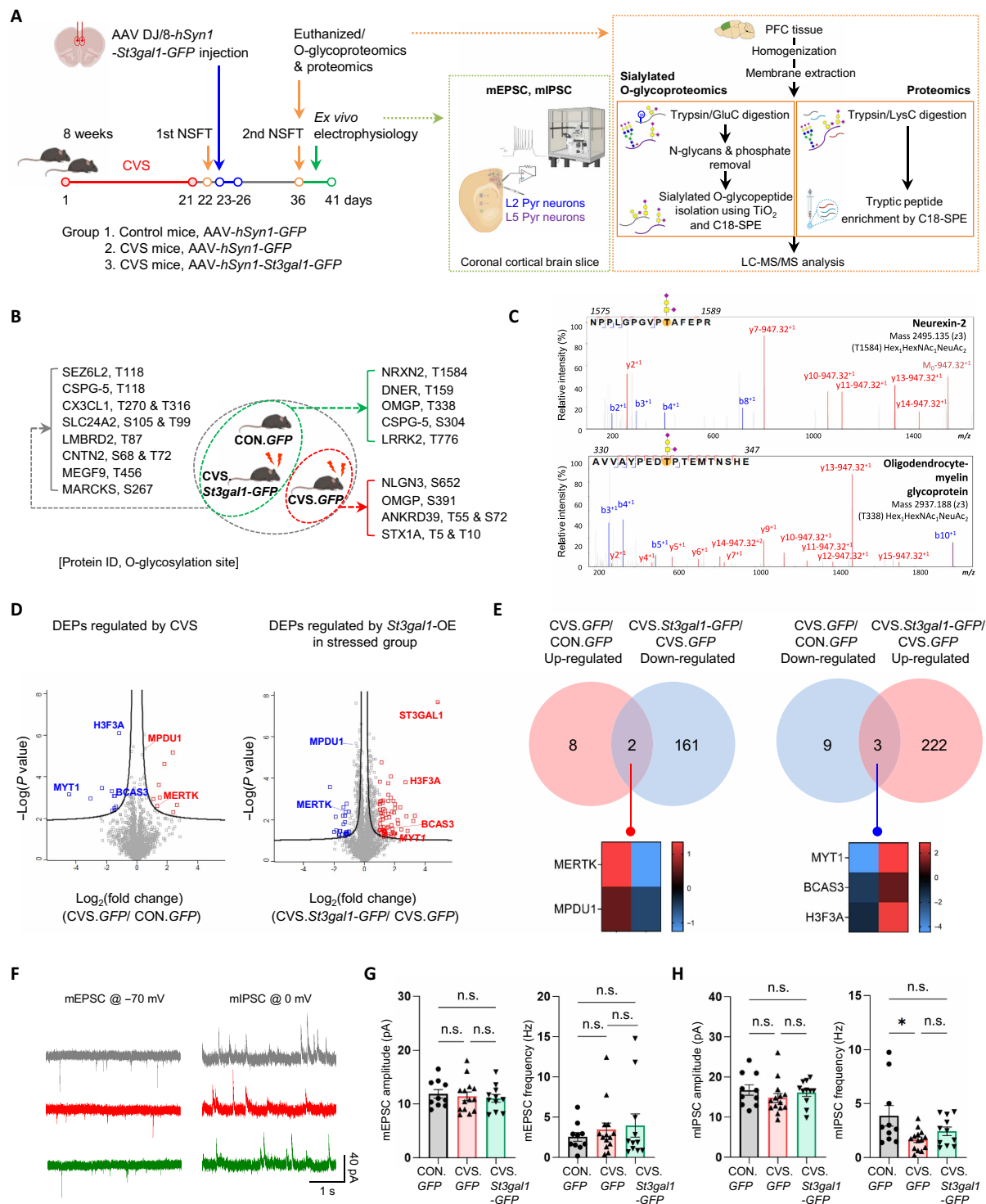


Fig. 6. ST3GAL1-driven sialylated O-glycoproteins, along with DEPs, are identified in the mPFC of gene-overexpressing CVS mice. (A) Streamlined workflow of sialylated O-glycoproteomic analysis, proteomic analysis, and ex vivo electrophysiology in CON-GFP, CVS-GFP, and CVS.*St3gal1*-GFP mouse groups. Created in BioRender. Chang, B. (2025) <https://BioRender.com/n77k1xg>. (B) List of proteins having sialylated O-glycosylation identified in the three groups. O-glycosylation was found at specific threonine or serine sites in each protein. (C) O-glycosylated site of neurexin-2 (top) and oligodendrocyte-myelin glycoprotein (bottom) identified by HCD-MS/MS analysis. (D) Volcano plot displaying the differential expression of mPFC proteins between two different mouse groups. An FDR threshold of 5% is denoted by black lines. Proteins with $\log_2(\text{fold change}) > 1$ are colored. (E) Venn diagram of up-regulated and down-regulated proteins in two different mouse groups. Expression of the five proteins shown in overlapping regions was restored by *St3gal1* overexpression in CVS mice. (F) Representative traces of mEPSCs and mIPSCs. (G) mEPSC amplitude (left) and mEPSC frequency (right). CON.GFP ($n = 10$), CVS.GFP ($n = 14$), and CVS.*St3gal1*-GFP ($n = 11$). (H) mIPSC amplitude (left) and mIPSC frequency (right). One-way ANOVA with Bonferroni multiple comparisons test. * $P < 0.05$. n.s., not significant.

responses of pyramidal neurons in the mPFC using ex vivo electrophysiology (Fig. 6, F to H, and fig. S15) (32, 33). No significant differences were observed among the groups (CON.*GFP*, CVS.*GFP*, and CVS.*St3gal1-GFP*) in miniature excitatory postsynaptic current (mEPSC) amplitude or frequency, miniature inhibitory postsynaptic current (mIPSC) amplitude, excitatory/inhibitory (E/I) ratio, resting membrane potential, input resistance, rheobase, paired-pulse ratio, or AMPA/NMDA (*N*-methyl-D-aspartate) ratio. However, mIPSC frequency was significantly reduced in the CVS.*GFP* group compared to CON.*GFP* but normalized in CVS.*St3gal1-GFP* mice, suggesting a potential modulatory role of *St3gal1* in inhibitory synaptic function (Fig. 6H).

In conclusion, chronic stress-induced alterations in *St3gal1* expression in the mPFC appear to regulate downstream O-glycoproteins, such as NRXN2, by modulating their stability or function, thereby influencing neuronal activity and downstream proteins in the mPFC. *St3gal1* overexpression not only reverses these molecular changes but also mitigates depressive-like behaviors, highlighting its potential as a therapeutic target.

DISCUSSION

Using an unbiased glycomics analysis across nine distinct brain regions, we identified a previously unknown mechanistic link between chronic stress-induced depression and region-specific O-glycosylation in the brain. Our findings reveal that CVS profoundly alters ST3GAL1-mediated sialylation of brain O-glycans, particularly in the PFC, a region strongly associated with depressive behaviors and related neuropathologies (23). These results enhance our understanding of the biochemical mechanisms underlying depression in the brain and highlight the promise of glycosylation-related pathways as previously unknown therapeutic targets for stress-induced psychiatric disorders.

Brain region-specific responses in O-glycosylation to chronic stress

A key finding from our depression study was the regional specificity of O-glycosylation alteration in the brain, with prominent changes observed in the CC, DIE, OLB, and PFC. The functional relevance of these regions to depression provides deeper insights into the neurobiological alterations induced by stress. Altered connectivity in the CC has been implicated in depression, particularly in disrupting the balance between emotional and rational processing (34). The DIE, encompassing the thalamus and hypothalamus, plays a pivotal role in stress signal processing and neuroendocrine regulation (35). Specifically, the hypothalamus, as part of the hypothalamic-pituitary-adrenal (HPA) axis, is central to the body's stress response, whereas the thalamus acts as a relay for sensory and emotional information (35, 36). Although previous research has linked reduced OLB volume and impaired olfactory function to MDD, the involvement of the OLB in depressive states remains underexplored, potentially connecting olfactory deficits to mood disorders—an exciting avenue for future research (37). Furthermore, an intriguing observation was the opposite patterns of O-glycosylation changes in the OLB during depressive states, which opens up new possibilities for identifying biomarkers and developing therapies. The PFC, a well-established region implicated in mood regulation, executive function, and stress responses, emerged as a critical focus of our study. Dysfunction in the PFC, including reduced synaptic plasticity and altered neuronal

signaling, is commonly associated with depression (38). Notably, the PFC plays a central role in connectivity with other key brain regions examined here, including the CC, DIE, and OLB (39–41). This integrative network highlights the PFC's pivotal role in coordinating emotional and cognitive processes under stress. Therefore, we prioritized the PFC as a focal point for further investigation. Other regions traditionally associated with depression, such as the HIP, STR, and cerebellum (CB) (39, 42), showed no notable O-glycosylation alterations. These findings underscore the spatial specificity and dynamic nature of O-glycosylation in the brain's response to chronic stress. Furthermore, this selective vulnerability suggests potential region-specific metabolic or cellular mechanisms underlying stress responses, emphasizing the distinct role of O-glycosylation within specific neural networks.

The cause-and-effect relationship between *St3gal1*-mediated O-glycan sialylation and depression

Differential expression of *St3gal1*, which encodes the sialyltransferase that adds Gal-NeuNAc to O-glycans, was most pronounced in the CC, OLB, and PFC—the same regions that exhibited the strongest stress-induced changes in sialylated O-glycans. *ST3GAL1* has been linked to bipolar disorder risk and to the severity of negative symptoms in schizophrenia (43, 44). In addition, the Mental Component Summary (MCS) score derived from studies of bipolar disorder, schizophrenia, and MDD is highly associated with *ST3GAL1* expression, which is decreased in individuals carrying the GWAS-identified SNP rs2736871, underscoring its clinical relevance (16). Our preclinical data in a mouse model support a functional role for this enzyme in mood regulation. Knockdown of *St3gal1* in the mPFC of unstressed mice induced depressive-like behaviors, including exaggerated novelty-induced feeding suppression and anhedonia, without affecting general locomotion. Conversely, overexpression of *St3gal1* in the mPFC reversed stress-induced behavioral deficits, reducing novelty-induced feeding suppression and promoting active coping behavior. Together, these findings indicate that down-regulation of *St3gal1* contributes causally to core depressive symptoms, whereas up-regulation confers resilience to chronic stress.

The mechanism by which *St3gal1* in the mPFC modulates depressive-like behaviors remains unclear. Our bulk RNA-seq analysis revealed significant enrichment of genes involved in KEGG pathways related to axon guidance and neurodegenerative diseases, including AD, PD, and HD. Brain sialylation has been extensively studied in the context of neurodegenerative conditions (45–47). In AD, altered sialylation of N- and O-glycans affects microglial function, protein aggregation, and synaptic plasticity in a brain region-specific and stage-specific manner (45, 46, 48). In PD, excessive sialylation of O-glycans and gangliosides has been associated with impaired immune recognition and increased microglial activation (47, 49, 50). These parallels suggest that ST3GAL1-dependent sialylation may represent a shared molecular pathway linking depression and neurodegeneration. Given that the mPFC is involved in goal-directed behavior, cognitive flexibility, and emotional regulation, future studies should explore whether ST3GAL1-mediated sialylation modulates these cognitive domains and contributes to a broader pathophysiological axis across neuropsychiatric and neurodegenerative disorders. Together, we believe that in-depth clarification of the upstream mechanisms by which chronic stress down-regulates *St3gal1*, as well as identification of interventions that selectively enhance its expression, may offer promising therapeutic

avenues for MDD and potentially other sialylation-associated neuropsychiatric and neurodegenerative diseases.

Sexual dimorphism in brain O-glycosylation patterns under chronic stress

A further intriguing observation was the sexual dimorphism in glycosylation changes. Although both male and female mice displayed depressive-like behaviors following CVS, alterations in O-glycosylation—particularly in the expression of the sialyltransferase gene *St3gal1*—were evident only in males. This suggests that stress-induced depression may involve distinct molecular mechanisms in male and female brains, underscoring the importance of investigating sex-specific pathways in stress and glycosylation biology. Collectively, our results underscore the need for further studies examining the sex-specific and region-specific roles of O-glycosylation in stress-related disorders.

St3gal1-mediated O-glycosylation on NRXN2 and GABAergic synapses in depression

Although we establish a causal relationship between *St3gal1* expression and these behavioral outcomes, the precise mechanisms—particularly how ST3GAL1 influences local circuit dynamics within the mPFC—remain to be fully elucidated. Our electrophysiological data show that chronic stress significantly reduces the frequency of mIPSCs in the mPFC, a deficit that is reversed by *St3gal1* overexpression. Given that *St3gal1* is primarily expressed in glutamatergic neurons (51), it is not immediately clear how it modulates presynaptic GABA release, as suggested by changes in mIPSC frequency. One plausible explanation is that ST3GAL1 indirectly regulates GABAergic inhibitory tone through a transsynaptic mechanism. We hypothesize that this modulation is mediated via O-glycosylation of NRXN2, a presynaptic adhesion molecule essential for the development and maintenance of GABAergic synapses. NRXN2 is expressed in both glutamatergic neurons and GABAergic interneurons (51). Our glycoproteomic analyses identified NRXN2 as a direct target of ST3GAL1-mediated sialylation. Under chronic stress, NRXN2 glycosylation is reduced, potentially destabilizing GABAergic synaptic architecture and diminishing inhibitory input, resulting in an E/I imbalance. Overexpression of *St3gal1* restores NRXN2 glycosylation, which may destabilize its activity and stabilize GABAergic synapses and reestablish the E/I balance within mPFC circuits—a critical determinant of mood, anxiety, and stress-related behaviors (52). Furthermore, our findings suggest a mechanistic convergence between *St3gal1* function and ketamine's antidepressant effects. Ketamine has previously been shown to restore chronic stress-induced GABAergic inhibition in the mPFC (53), and our data indicate that it also prevents the stress-induced down-regulation of *St3gal1*. These observations support the hypothesis that both interventions may act through a shared ST3GAL1-NRXN2-GABAergic synapse axis, representing a previously unrecognized molecular and circuit-level pathway underlying their antidepressant actions. This model offers a compelling framework for understanding how O-glycosylation—a previously underappreciated PTM—can regulate synaptic function and emotional behavior through cell type-specific and transsynaptic mechanisms. However, we cannot rule out the involvement of vasoactive intestinal peptide (VIP)-positive neurons in this process as *St3gal1* is expressed in these cell types as well (51). Future studies, such as cell type-specific manipulation of *St3gal1* and site-directed mutagenesis of NRXN2, particularly those targeting the sialylated

O-glycosylation site threonine-1584 (T1584), will be essential to validate this mechanism and further explore the therapeutic potential of targeting this pathway.

Downstream molecular networks modulated by *St3gal1* in the mPFC in depression

In addition to NRXN2, MERTK, a downstream target whose expression is reversed by *St3gal1* overexpression, appears to play a role in regulating stress responses and synaptic plasticity, suggesting that it may be another key player in chronic stress-induced depressive-like behaviors (54–56). Our network analysis revealed associations among proteins directly or indirectly modulated by ST3GAL1, with NRXN2 acting as a core node. Thus, future studies on how ST3GAL1 affects potential glycoproteins and other downstream targets, particularly those with reversible expression changes in the PFC, will enhance our understanding of the molecular pathways influenced by O-glycosylation.

Although we identified several interesting targets regulated by *St3gal1* overexpression, for both proteomics and glycoproteomics analyses, we focused on proteins or glycoproteins significantly altered in the CVS.GFP group compared to the CON.GFP group. Among the proteins altered by chronic stress, we then examined those whose expression was reversed by *St3gal1* overexpression (CVS.*St3gal1*-GFP versus CVS.GFP), aiming to identify prefrontal proteins affected by stress and potentially normalized through increased *St3gal1* expression. Given that our analysis was qualitative rather than quantitative, the extent of *St3gal1* overexpression is unlikely to have influenced the presence or absence of sialylation on specific glycoproteins. Although proteomic and glycoproteomic data from CON.*St3gal1*-GFP mice were not included in this study, behavioral comparisons between CON.GFP and CON.*St3gal1*-GFP animals revealed no significant differences in anxiety or locomotor activity in the OFT nor in depressive-like behaviors assessed by the NSFT, TST, and SPT (fig. S15). These findings suggest that *St3gal1* overexpression alone does not influence baseline behavior. Nevertheless, we acknowledge that including proteomic and glycoproteomic analyses of control mice following *St3gal1* overexpression (CON.*St3gal1*-GFP) would further strengthen the validation of the glycosylation patterns observed in CVS.*St3gal1*-GFP mice.

In summary, our work highlights ST3GAL1-mediated O-glycosylation, acting through mPFC circuits, as a previously unrecognized and important pathway contributing to depressive-like behaviors. By revealing the region-specific and sex-dependent nature of these alterations, this study suggests previously underexplored molecular connections to depression and supports the consideration of glycosylation-related pathways as potential targets for future therapeutic development.

MATERIALS AND METHODS

Animals

Experiments used male and female C57BL/6J mice aged 8 to 14 weeks, sourced from the Jackson Laboratory (Bar Harbor, USA) and subsequently bred in-house for experimental use. The mice were housed in groups of five per cage under standard conditions (22° ± 5°C, 12-hour/12-hour light/dark cycle, lights on at 8:00 a.m.) without environmental enrichment. All procedures were performed during the light cycle and approved by the Institutional Animal Care and Use Committee (IACUC) of the Institute for Basic Science, Korea.

The study was conducted under protocol number IBS-2024-002, with all protocols designed to minimize animal discomfort in accordance with IACUC guidelines.

Animal model of depression

CVS was selected as a well-established model for inducing depressive-like behaviors in both male and female mice (57–59). CVS uses a sequence of three stressors that are less labor-intensive than those used in the mild unpredictable stress model yet still produce consistent and reliable depressive-like behaviors (fig. S1). We implemented a 3-week protocol to robustly induce these behaviors and to ensure conclusive biochemical outcomes. In this study, three different stressors—foot shock, tail suspension, and tube restraint—were introduced to 11-week-old adult mice repeatedly over 21 days. Each day, mice were exposed to one stressor for 1 hour. On the first day, adult mice were placed in an electric shocker and subjected to 100 random foot shocks at 0.45 mA over 1 hour. On the second day, the same mice were suspended by the tail for 1 hour. On the third day, mice were restrained in a 50-ml tube for 1 hour. This cycle was repeated throughout the 21-day period. Control mice were maintained under standard, nonstressed conditions for 3 weeks. Following the CVS protocol, depressive-like behavior was assessed using the NSFT, as described below. Mice were exposed to stressors between 14:00 and 16:00 (ZT 6 to 8) to avoid light stimulation during their active (lights-off) phase.

Multimomics

For subsequent biochemical analyses, including glycomics, glycoproteomics, and bulk RNA-seq, we used only the NSFT to assess depressive-like behavior to minimize stress exposure in control animals. To evaluate the effects of *St3gal1* knockdown or overexpression, we conducted a series of behavioral tests, as outlined in the schematic provided in each relevant figure.

Mouse brain O-glycomics

Right after the NSFT of adult male mice ($n = 10$ CON and 10 CVS), they were euthanized by cervical dislocation, and brain tissues were immediately dissected into nine subregions: CC, PFC, STR, HIP, OLB, DIE, midbrain (MB), pons-medulla (PM), and CB. All brain regions were obtained from both hemispheres. O-glycans of mouse brain tissues were released by following steps: membrane extraction, O-glycopeptide preparation, chemical release of O-glycans, and O-glycan enrichment. Briefly, each tissue was homogenized in a buffer containing 20 mM Hepes-KOH (pH 7.4), 0.25 M sucrose, and protease inhibitors (Sigma-Aldrich, 11873580001) and then sonicated on ice. Lysates (2 mg of protein) were centrifuged at 43,000 rpm for 80 min at 4°C, resuspended in 0.2 M sodium carbonate (pH 11), and washed with 50 mM ammonium bicarbonate (ABC) (pH 7.5). Membrane pellets were solubilized in 100 mM ABC with dithiothreitol, denatured at 60°C for 50 min, and alkylated with iodoacetamide for 40 min in the dark. Trypsin (Promega, V5111) (1:100) was added, and samples were incubated overnight at 37°C. Trypsin activity was halted by heating at 95°C for 5 min. N-glycans were removed using peptide N-glycosidase F (PNGase F) (NEB, P0705L), and O-glycopeptides were enriched using C18-based solid phase extraction (SPE) (Waters, 186000308), with peptides eluted using 20 and 40% isopropanol (IPA) in water (v/v) and 100% IPA. Tryptic O-glycopeptides underwent β -elimination in an alkaline solution (1.0 M sodium borohydride and 0.1 M sodium hydroxide) at 45°C for 16 hours. After quenching with 1.0 M hydrochloric acid solution,

O-glycans were purified using porous graphitized carbon (PGC)–SPE (Thermo Fisher Scientific, 60106-302) and eluted with 20 and 40% acetonitrile (ACN) in water containing 0.05% trifluoroacetic acid (TFA). To further remove small peptides and remained salts, O-glycan samples of brain tissues were repurified using iSPE-Hydrophilic Interaction Chromatography cartridges (HILICON, 200.001.0100). The cartridge was washed and conditioned according to the protocol previously reported (60). The dried O-glycan sample was reconstituted in 90% ACN in water with 1% TFA (v/v) and was loaded onto the cartridge. The purified O-glycans were collected with a solution of 0.1% TFA in water (v/v) and then dried under vacuum.

O-glycans were separated on a Hypercarb column using Ultimate 3000 UHPLC (Thermo Fisher Scientific) coupled with a TIMS Q-TOF MS (Bruker Daltonics Inc.) system. All O-glycans were separated through the Hypercarb column (Thermo Fisher Scientific, 35003-102130). The LC gradient condition for O-glycan separation was optimized using mobile phases of (A) 100% H₂O with 0.1% formic acid (FA) (v/v) and (B) 100% ACN with 0.1% FA (v/v). MS spectra were acquired in positive ion mode over a mass-to-charge (m/z) range of 300 to 2000, and CID MS/MS was performed with collision energies tailored to precursor ion m/z values. LC-MS analysis of all O-glycan samples was performed in triplicate. Raw LC-MS and MS/MS data were processed using Bruker Compass Data Analysis 6.1. MS peaks of all ion species were filtered on the basis of expected isotopic distribution with a signal-to-noise ratio of 3.0. O-glycan compositions were identified by accurate mass [≤ 20 ppm (parts per million)] and validated by CID MS/MS. The peak area of each O-glycan ion was indicated in the EIC, which normalized against the sum of all ion peaks of O-glycans for relative quantitation.

Sialylated O-glycoproteomics of the mouse brain

Sialylated O-glycopeptides of PFC tissues were selectively enriched by four steps: membrane extraction, O-glycopeptide release, removal of N-glycosylation and phosphorylation, and selective capture of sialylated O-glycopeptides using titanium dioxide (TiO₂). PFC lysates underwent membrane extraction as described in the O-glycan isolation section, followed by dissolution in 0.2% RapiGest (w/v) in 100 mM ABC. Proteins were denatured with tris(2-carboxyethyl) phosphine hydrochloride (TCEP) for 30 min and alkylated with chloroacetamide (CAA) for 40 min at room temperature. After digestion with Glu C (Promega, V1651) overnight at 37°C at an enzyme to protein ratio of 1:100 and heating at 98°C for 4 min, trypsin digestion (1:100) was performed for 8 hours at 37°C, with subsequent heat inactivation. To capture sialylated O-glycopeptides, PNGase F and Protein Phosphatase (NEB, P0753S) were used to remove N-glycans and phosphates. Samples were purified using C18-SPE, with peptides eluted in sequence using 20, 40, and 100% IPA. Sialylated O-glycopeptides were enriched with TiO₂ spin tips as per an established protocol (61, 62). Dried samples were reconstituted in a loading buffer (80% ACN, 5% TFA, and 1 M glycolic acid), loaded onto TiO₂ tips, and washed with buffers [80% ACN in water with 1% TFA (v/v) and 20% ACN in water with 0.2% TFA (v/v)] to remove nonsialylated components. Elution was achieved with ammonia water (pH 11), followed by C18-SPE purification. The buffer solutions used here were as follows: loading and washing buffer solution [100% water with 0.2% TFA (v/v)] and eluting buffer solutions [80% ACN in water and 80% ACN in water with 0.2% TFA (v/v)]. Samples were vacuum dried and stored at –80°C for analysis.

Sialylated O-glycopeptide samples from three mice were pooled for LC-MS/MS analysis using an UltiMate 3000 RSLCnano and Orbitrap Q Exactive (Thermo Fisher Scientific) system. Peptides were captured in a C18 trapping column (Thermo Fisher Scientific, PN164535) and separated using a C18 analytical column (Thermo Fisher Scientific, PNES903) under a nanoLC gradient of mobile phases A (0.1% FA in water) and B (0.1% FA in ACN). The gradient conditions applied were as follows: 5% B (0 to 15 min), 5 to 20% B (15 to 80 min), 20 to 40% B (80 to 95 min), and 40 to 95% B (95 to 98 min) at 0.3 μ l/min. HCD-MS/MS was conducted in positive ion mode over the m/z range of 400 to 2000, with collision energies adjusted for precursor ion masses. Each sample was analyzed in triplicate. Raw LC-MS/MS data were processed with MetaMorpheus 1.0.6 using the *Mus musculus* proteome database.

Proteomics of PFC tissues

Following ultracentrifuge-based membrane extraction, samples were dissolved in 0.2% Rapigest in 100 mM ABC. Proteins were denatured with 5 mM TCEP and alkylated with 20 mM CAA at room temperature. A Trypsin/Lys-C protease mix (Promega, V5071) was added an enzyme to protein ratio of 1:30, and samples were incubated overnight at 37°C. Peptides were purified and concentrated using C18-SPE with a loading buffer (100% water with 0.2% TFA) and eluting buffers (80% ACN in water or with 0.2% TFA). Dried peptide samples were reconstituted in 0.1% FA and prepared for LC-MS/MS analysis.

Proteomic analysis was performed using a nanoLC-MS/MS system (UltiMate 3000 RSLCnano coupled to Orbitrap Q Exactive). Peptides were initially trapped on a C18 column and separated using a C18 analytical column with the same nanoLC gradient and mobile phase conditions as the sialylated O-glycoproteomic analysis. Raw LC-MS/MS data from triplicate runs were analyzed using MaxQuant with the UniProt mouse database. Search parameters included fixed cysteine carbamidomethylation, as well as variable modifications of methionine oxidation. Peptides with a minimum length of six amino acids and a false discovery rate (FDR) of 0.05% were considered for protein identification, requiring at least two unique peptides per protein group. The label-free quantification (LFQ) algorithm in MaxQuant was used for protein quantification.

Transcriptomics

RNA purification and sequencing

Right after the NSFT of adult male mice ($n = 3$ CON and $n = 3$ CVS), the mouse brain was dissected and RNA from each subregion was extracted and purified using the Maxwell RSC system and RNA extraction kit (Promega, AS1340) in accordance with the manufacturer's protocol. The purified RNA was eluted in nuclease-free water, and its concentration was determined with a NanoDrop UV-Vis spectrophotometer (Thermo Fisher Scientific, NanoDrop One). RNA quality was evaluated using a Bioanalyzer 2100 (Agilent Technologies, Wilmington, DE, USA). RNA library preparation and sequencing were performed by Inocras (San Diego, CA, USA) using the KAPA mRNA HyperPrep Kit (Illumina, San Diego, CA, USA) to generate high-quality RNA-seq libraries. Sequencing was conducted on an Illumina NovaSeq 6000 platform (Illumina, San Diego, CA, USA).

RNA-seq data analysis

RNA-seq data analysis was carried out using Partek Flow (version 12.2.2). Initially, prealignment quality control was performed to ensure high-quality reads. The reads were then aligned to the mouse genome assembly (mm39, Ensembl transcripts release 98) using the

STAR aligner (version 2.7.8a) and normalized to transcripts per million (TPM). Gene set analysis (GSA) was used to identify DEGs with a P value of <0.05 between the two groups. The DEGs were further analyzed through KEGG pathway and GO enrichment analyses.

Quantitative real-time reverse transcription PCR

O-glycosylation genes up-regulated and down-regulated in RNA-seq were verified through RT-qPCR. The PCR reaction was performed using the following primers and TaqMan probes (Integrated DNA Technologies): *St3gal1* (Mm.PT.58.16698772, primer 1: CCT-CTCCATCATCTTCTCCATTC; primer 2: GTGCCAATTTCTTTT-GCTGTC); *Gcnt4* (Mm.PT.58.45858596, primer 1: GCAGGCGGA-ATAGAGATCTAAG; primer 2: GCTGGAGAGTGTATTTAAAGC-AAC); *Fut9* (Mm.PT.58.32573215, primer 1: ACTCACGCATCTGATAGCAG; primer 2: GACGATTAGAAATGGGCGAAGA); *B3gal5* (Mm.PT.58.43397113, primer 1: CAGCGATCTTAGGAAGTGAG-AA; primer 2: ATGATGATTGTGTCTCTCGGAA); and *Gapdh* (Mm.PT.39a.1, primer 1: AATGGTGAAGGTCGGTGTG; primer 2: GTGGAGTCATACTGGAACATGTAG). Gene expression levels were calculated by the $2^{-\Delta\Delta CT}$ method, and the levels of all glycosylation genes were normalized to housekeeping gene *Gapdh*.

AAV vectors

AAV vectors and validation

AAVs were used here to generate adult mouse models with gene knockdown and overexpression of *St3gal1*. AAV-pSicoR-*St3gal1*.shRNA-mCherry, AAV-pSicoR-scramble.shRNA-mCherry, AAV-*hSyn1-St3gal1-GFP*, and AAV-*hSyn1-GFP* vectors from serotypes DJ/8 were obtained from the virus facility in the Center for Cognition and Sociality, Institute for Basic Science. All viral vectors were tested in the primary culture of mouse cortical neurons (30). Mouse neurons were isolated from the brain of C57BL/6J mouse embryos and plated at a density of 1×10^6 cells per well in the 24-well plate. After growing the cells for 7 days, each viral vector was added to a well at equal titers (1×10^{12} genome copies/ml) in both control and knockdown (or overexpression) groups. The efficiency of *St3gal1* knockdown or overexpression by treatments of AAV vectors was assessed by RT-qPCR analysis. The neurons were collected on day 5, and their RNAs were extracted and purified by a Maxwell RSC instrument and RNA extraction kit.

Virus surgery

For the gene knockdown model, 11-week-old C57BL/6J mice were anesthetized with an intraperitoneal injection of ketamine (120 mg/kg) and xylazine (9.8 mg/kg). AAV-*St3gal1.shRNA* or AAV-scramble vectors were then injected bilaterally into the mPFC. For the overexpression experiment, AAV-*hSyn1-St3gal1-GFP* or AAV-*hSyn1-GFP* vectors were injected into the mPFC of 11-week-old CVS mice following respiratory anesthesia with isoflurane. Ten days after viral vector injection, transduction efficiency was assessed by RT-qPCR of *St3gal1*. The injection coordinates were anteroposterior [AP (+1.7)], mediolateral [ML (± 0.35)], and dorsoventral [DV (2.15 mm from the skull)].

Ketamine administration

Male C57BL/6J mice (11 weeks old) underwent 21 days of CVS as described above. Following the NSFT to assess depressive-like behaviors, one CVS group ($n = 4$) received ketamine hydrochloride [10 mg/kg, intraperitoneally (ip)] (Yuhan Co., South Korea) on days 23 and 26, whereas another CVS group ($n = 4$) and the control group ($n = 4$) received saline. The NSFT was performed 24 hours after

each injection of saline or ketamine. Normal male C57BL/6J mice (8 weeks old) also received intraperitoneal injections of ketamine hydrochloride (10 mg/kg, ip) (CON.Ketamine group, $n = 7$) or saline (CON.Saline group, $n = 7$) on days 1 and 4.

Immunohistochemistry and confocal imaging

For histology, mice were anesthetized using isoflurane, and the brains were collected following transcardiac perfusion. The brains were postfixed for 24 hours in a 4% paraformaldehyde solution at 4°C. Coronal sections of the fixed brains were sliced at 40 μm in thickness and thoroughly washed in 0.1 M phosphate-buffered saline (PBS). The slices were permeabilized with 0.3% Triton X-100 for 30 min and blocked with 5% normal goat serum (NGS) for an hour at room temperature. The slices were then incubated overnight with specific primary antibodies [rabbit anti-ST3GAL1 antibody (1:200, Invitrogen, PA5-21721) and mouse anti-NeuN antibody (1:250, Merck Millipore, MAB377)] at 4°C. After washing with 0.1 M PBS, the slices were incubated with secondary antibodies (donkey anti-mouse Alexa Fluor 647 and goat anti-rabbit Alexa Fluor 594; the Jackson Laboratory) for 2 hours at room temperature. Following thorough washing with 0.1 M PBS, the tissue slices were mounted onto glass slides and coverslipped with a mounting solution containing DAPI (Vector Laboratories, H-1200-10). Fluorescence imaging was performed using a Zeiss LSM900 confocal microscope (Carl Zeiss), and the MFI was quantified using ImageJ software. The nuclear overlap in a subset of micrographs reflects background signal, which became visible because detector gain was increased to visualize the very low staining in CVS.GFP sections; the strong perinuclear-like signal in CVS.St3gal1-GFP potentially represents ST3GAL1 expression in ER (endoplasmic reticulum)/Golgi localization. To ensure valid comparisons, all comparative images were acquired under identical settings.

Ex vivo electrophysiology

Mice were deeply anesthetized with 1.5–2% isoflurane and decapitated, and then the brain was quickly excised from the skull and submerged in an ice-cold cutting solution: 5 mM KCl, 1.25 mM NaH_2PO_4 , 26 mM NaHCO_3 , 10 mM D-glucose, 0.5 mM CaCl_2 , 10 mM MgCl_2 , and 212.5 mM sucrose, bubbled with 95% O_2 and 5% CO_2 (pH 7.4). The brain was glued onto the stage of a vibrating microtome (D.S.K, LinearSlicer PRO7N, Japan). Coronal cortical brain slices were cut into 300- μm -thick sections and transferred to an extracellular aCSF solution: 126 mM NaCl, 24 mM NaHCO_3 , 1 mM NaH_2PO_4 , 2.5 mM KCl, 2.5 mM CaCl_2 , 2 mM MgCl_2 , and 10 mM D-(+)-glucose (pH 7.4). Slices were incubated at room temperature for at least 30 min before recording and then transferred to a recording chamber continuously perfused with a CSF solution using a peristaltic pump (Minipuls 3, Gilson) at a rate of 2–3 ml/min. Cells were visualized using a 40x Nikon objective (NIR Apo, 40/0.8w, DIC N2, $\infty/0$ WD 3.5) equipped with infrared differential interference contrast (IR-DIC) optics. Whole-cell recordings were made from layer V pyramidal neurons in the prelimbic cortex. For neuronal firing activity, pipette resistance was typically 5 to 7 megohms and the pipette was filled with an internal solution (in mM): 120 mM K-gluconate, 20 mM KCl, 2 mM NaCl, 20 mM Hepes, 0.5 mM EGTA, 10 mM glucose, 2 mM Mg-ATP, and 0.5 mM Na_2 -GTP, pH adjusted to 7.3 with CsOH (280 to 290 mosmol). For miniature synaptic events, mEPSCs and mIPSCs, pyramidal neurons in layer V of the prelimbic cortex were held at -70 and 0 mV in the presence of

tetrodotoxin (TTX; 1 μM), respectively. The pipette was filled with an internal solution: 130 mM CsMESO₄, 10 mM Hepes, 10 mM EGTA, 9 mM TEA-Cl, 4 mM Mg-ATP, and 0.3 mM Na_2 -GTP, pH adjusted to 7.3 with CsOH (280 to 290 mosmol). Synaptic responses were evoked by 0.1-Hz stimulation on layer II/III (100- μs duration; 50–80- μA intensity) via a constant current isolation unit. The pipette was filled with an intracellular solution containing 135 mM CsMESO₄, 8 mM NaCl, 0.25 mM EGTA, 10 mM Hepes, 2 mM Mg-ATP, 0.5 mM Na_2 -GTP, and 5 mM QX-314, pH adjusted to 7.3 with CsOH (280 to 290 mosmol). Bicuculline (10 μM) was bath applied to block GABA_A (γ -aminobutyric acid type A) receptor-mediated currents. The evoked currents at -70 and 40 mV were identified as the AMPAR (AMPA receptor)-mediated and NMDAR (NMDA receptor)-mediated currents, respectively. The amplitude of the evoked EPSC at 50-ms poststimulus was considered as the NMDAR-mediated currents. Electrical signals were digitized and sampled at 50- μs intervals with a Digidata 1550B and Multiclamp 700B amplifier (Molecular Devices) using pCLAMP11.2 software. Data were filtered at 2 kHz.

Behavior studies

All behavioral tests were adapted from published protocols in previous studies (19, 63–66). Animals were randomly assigned to control and stressed groups. Sample size was determined using the resource equation ($E = N - B - T$), where N is the total number of animals, B is the number of experimental groups, and T is the number of treatments. We ensured that E fell within the recommended range of 10 to 20 (67). Behavioral tests were conducted primarily between 14:00 and 20:00 (ZT 6 to 12) to avoid light stimulation during the animals' active (lights-off) phase. All mice were habituated to the testing room for at least 1 hour before testing. Postbehavioral sampling was performed under blinded conditions.

Novelty-suppressed feeding test

Mice were food deprived for 24 hours before behavioral testing. Each mouse was placed in a brightly lit open field box (41 cm by 41 cm by 41 cm, white) with a single food pellet in the center. The mouse was positioned in a corner of the box and allowed to explore until it began biting the pellet. Immediately after the test, the mouse was returned to a corner of its home cage with a food pellet to measure food consumption under familiar conditions. When multiple NSFTs were conducted, the boxes varied in size and color (40 cm by 59 cm by 34 cm, gray or 41 cm by 41 cm by 41 cm, gray) to provide a novel environment. The latency to the first bite was capped at 15 min in the novel environment and 90 s in the home cage. The arena floor was not covered with bedding material.

Open field test

To observe locomotion and spontaneous activity of mice, they were allowed to freely explore an open field box (41 cm by 41 cm by 41 cm) under dim light conditions of 10 lux. One mouse was placed in a corner of the box. In the box, the total distance traveled by the mouse and the total time spent in the center and whole area were recorded with EthoVision XT video tracking software (Noldus Information Technology, Wageningen, The Netherlands) and the recording data for 5 min and total 30 min were analyzed separately.

Tail suspension test

Each mouse was suspended 30 cm above the floor by taping ~ 1 cm from the tip of the tail. The mice's movements were recorded for 6 min, and the immobility time during the last 4 min was measured to assess stress-coping behavior.

Sucrose preference test

The SPT was conducted to assess anhedonia toward a sweet gustatory stimulus. In the habituation period for testing, mice were given two water bottles for 24 hours, followed by two bottles filled with 1% sucrose solution for another 24 hours. After food and water deprivation for 24 hours, a mouse was given one bottle of water and one bottle containing 1% sucrose solution. The weight of both bottles was checked at the first hour, 4 hours, and 8 hours. The positions of two bottles were switched after 4 hours for an additional 4 hours. After testing during 8 hours, sucrose preference was assessed by determining the percentage of total sucrose consumption divided by total liquid consumption (water and 1% sucrose solution).

Statistical analysis

Statistical analyses were conducted using GraphPad Prism 10 (Dotmatics, San Diego, CA, USA). Depending on the experimental design, we used either a two-tailed unpaired *t* test (for comparisons between two groups) or one-way analysis of variance (ANOVA) followed by Bonferroni multiple comparisons test (for comparisons among more than two groups). Each point in data is one biological sample, and data are indicated as means $\pm$ SEM. In the figures of behavior tests, significance was indicated as the following *P* values: **P* < 0.05, ***P* < 0.01, ****P* < 0.001, and *****P* < 0.0001. Adjusted *P* values were used for multiple comparisons.

Supplementary Materials

The PDF file includes:

Figs. S1 to S15

Legends for tables S1 to S9

Other Supplementary Material for this manuscript includes the following:

Tables S1 to S9

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Abnormal O-glycan sialylation in the mPFC contributes to depressive-like behaviors in male mice

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